

CWA SCIENCE CLASSES

Concept With Abhishek | Class 12 — Maths

Chapter 1: Relations and Functions

1. Relation: A subset of the Cartesian product $A \times B$, i.e. a set of ordered pairs.
2. Types of Relations: Empty, Universal, Reflexive, Symmetric, Transitive and Equivalence relation.
3. Equivalence Relation: A relation that is reflexive, symmetric and transitive simultaneously — partitions a set into equivalence classes.
4. Function: A special relation where every element of domain has exactly one image in codomain.
5. Types of Functions: One-one (injective), Onto (surjective), Bijective (both one-one and onto), Many-one, Into.
6. Composition of Functions: $(f \circ g)(x) = f(g(x))$. Composition is associative but not commutative in general.
7. Invertible Function: A function is invertible if and only if it is bijective (one-one and onto).